

Use of artificial neural network black-box modeling for the prediction of wastewater treatment plants performance

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Abstract

A reliable model for any wastewater treatment plant is essential in order to provide a tool for predicting its performance and to form a basis for controlling the operation of the process. This would minimize the operation costs and assess the stability of environmental balance. This process is complex and attains a high degree of nonlinearity due to the presence of bio-organic constituents that are difficult to model using mechanistic approaches. Predicting the plant operational parameters using conventional experimental techniques is also a time consuming step and is an obstacle in the way of efficient control of such processes. In this work, an artificial neural network (ANN) black-box modeling approach was used to acquire the knowledge base of a real wastewater plant and then used as a process model. The study signifies that the ANNs are capable of capturing the plant operation characteristics with a good degree of accuracy. A computer model is developed that incorporates the trained ANN plant model. The developed program is implemented and validated using plant-scale data obtained from a local wastewater treatment plant, namely the Doha West wastewater treatment plant (WWTP). It is used as a valuable performance assessment tool for plant operators and decision makers. The ANN model provided accurate predictions of the effluent stream, in terms of biological oxygen demand (BOD), chemical oxygen demand (COD) and total suspended solids (TSS) when using COD as an input in the crude supply stream. It can be said that the ANN predictions based on three crude supply inputs together, namely BOD, COD and TSS, resulted in better ANN predictions when using only one crude supply input. Graphical user interface representation of the ANN for the Doha West WWTP data is performed and presented.

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1. Introduction

The increased concern about environmental issues has encouraged specialists to focus their attention on the proper operation and control of wastewater treatment plants (WWTPs). The characteristics of influent to the WWTPs are varied from one plant to another depending on the type of community lifestyle. Therefore, the performance of any WWTP depends mainly on local experience of a process engineer who identifies certain states of the plant (Hong et al., 2003). The type of influent for any plant is also time-dependent and it is difficult to

have a homogeneous influent to a WWTP (Hamed et al., 2004). This may result in an operational risk impact on the plant. Serious environmental and public health problems may result from improper operation of a WWTP, as discharging contaminated effluent to a receiving water body can cause or spread various diseases to human beings. Accordingly, environmental regulations set restrictions on the quality of effluent that must be met by any WWTP.

Safer operation and control of a WWTP can be achieved by developing a modeling tool for predicting the plant performance based on past observations of certain key product quality parameters. Wastewater treatment plants involve several complex physical, biological and chemical processes. Often these processes exhibit non-linear behaviors which are difficult to describe by linear mathematical models (Plazl et al., 1999). In addition, the variability of the influent characteristics, in terms of composition, strength

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and flow rates, might influence model parameters, and consequently operational control, significantly (Hamoda et al., 1999). Therefore, modeling a WWTP is a difficult task and most of the available models are just approximations based on, probably severe, assumptions (Lee and Park, 1999; Cote et al., 1995; Hamoda et al., 1999; Plazl et al., 1999; Bhat et al., 1990).

Owing to their high accuracy, adequacy and quite promising applications in engineering (Govindaraju, 2000; Maier and Dandy, 2000; Neelakantan et al., 2001), artificial neural networks (ANNs) can be used for modeling such WWTP processes. The ANN can be used for better prediction of the process performance. It normally relies on representative historical data of the process. In a wastewater treatment plant, there are certain key parameters which can be used to assess the plant performance. These parameters could include biological oxygen demand (BOD), suspended solid (SS) and chemical oxygen demand (COD). Most of the available literature on the application of ANNs for modeling WWTPs utilized these parameters. For example, Oliveira-Esquerre et al. (2002) obtained satisfactory predictions of the BOD in the output stream of a local biological wastewater treatment plant for the pulp and paper industry in Brazil. The principle component analysis was used to preprocess the data in the back propagated neural network. The Kohonen Self-Organizing Feature Maps (KSOFM) neural network was applied by Hong et al. (2003) to analyze the multidimensional process data and to diagnose the inter-relationship of the process variables in a real activated sludge process. The authors concluded that the KSOFM technique provides an effective analysis and diagnostic tool to understand the system behavior and to extract knowledge from multidimensional data of a large-scale WWTP. Hamed et al. (2004) developed ANN models to predict the performance of a WWTP based on past information. They used such data as BOD, COD and SS collected from a conventional treatment plant in Cairo, Egypt. The authors found that the ANN-based models provide an efficient and robust tool in predicting WWTP performance. Zhu et al. (1998) proposed a time-delay neural network (TDNN) modeling method for predicting the effectiveness of a biological treatment process in terms of BOD measurements. The simulation results of the authors using real process data showed that the one-line training of the neural network model resulted in an improvement in the prediction accuracy. A procedure has been developed by Cote et al. (1995) using a neural network to improve the accuracy of an existing mechanistic model of the activated sludge process, previously described by Lessard and Beck (1993). A hybrid neural network approach, which combines mechanistic and neural network models, has also been used to model a full-scale coke-plant wastewater treatment process (Lee et al., 2002). The results of the authors indicated that the parallel hybrid neural modeling approach is a useful tool for accurate and cost-effective modeling of biochemical process, in the absence of other reasonably accurate process models.

In this work, ANN models were developed to predict the performance of a wastewater treatment plant based on the available historical data. The ANN-based models were applied to one of the wastewater treatment plants in the State of Qatar, Gulf zone, namely the Doha West WWTP. The models were tested for different configurations of input–output data collected from different locations of the plant. Using the results of this modeling process, the plant operator will be able to have an assessment of the expected plant effluent for a given quality of the wastewater stream at input locations.

2. Plant layout: a case study

The developed ANN model was applied to the Doha West WWTP which is considered a relatively new and consistent plant compared to the other WWTP in the state of Qatar. The Doha West WWTP is essentially designed to handle the following loads: population served = 200,000; average dry weather flow = 54,000 m³/d; peak flow = 183,000 m³/d; BOD₅ of the raw WW = 300 mg/l; total suspended solids (TSS) of the raw WW = 300 mg/l.

A schematic diagram of the plant is shown in Fig. 1. The crude sewage (CS) from different pumping stations is collected and screened for floating debris and removal of grit is carried out by the grit collector and grit elevators. Primary settlement tanks (PST) are utilized to settle 65–75% of the solids. Settled solids are scrapped down in the hoppers of the PST with the help of mechanical drive scrappers. These settled solids are removed by the Hydro Valves which open in the Consolidation Sludge Tank. Aerobic bacteria are activated by aeration and mixing with activated sludge to reduce the volume of mixed liquor. Primary treated effluent is mixed with the returned activated sludge from the secondary settlement tank and uniformly distributed in channels for aeration with the help of mechanically driven aerators. Mixed liquor out of the aeration tank is made to settle in the secondary settlement tanks. In the post-treatment, the secondary treated effluent is pre-chlorinated and lifted by screw pumps for uniform distribution to sand filters. The resulting stream, designated as final effluent (FE), flows down into the wet well.

3. Data collection

The available data for the Doha West WWTP were carefully investigated. It was decided to relate the outputs of the secondary treatment effluent (STE) stream to the inputs of the CS stream. This is because the outputs of the FE stream were almost similar to that of the STE stream and more comprehensive data was available for the STE stream. Therefore, measurements of the BOD, COD, and TSS in the CS stream and STE stream were collected over a one-year period. This period was satisfactory as it covers all probable seasonal variations in the studied variables. The measurements were performed in the plant almost every 5 days. These data are shown in Fig. 2. The

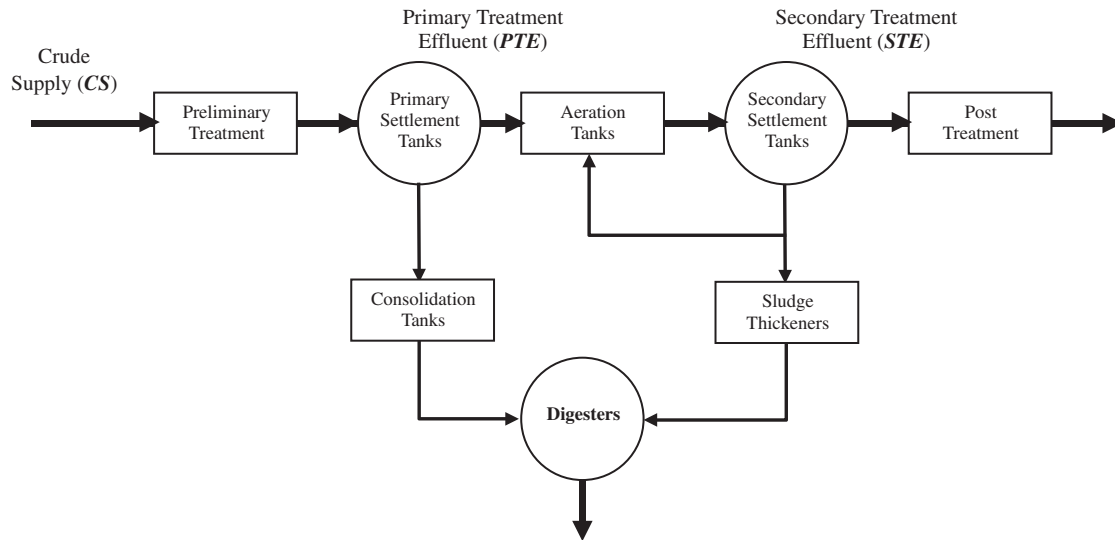


Fig. 1. Schematic diagram of the Doha West WWTP.

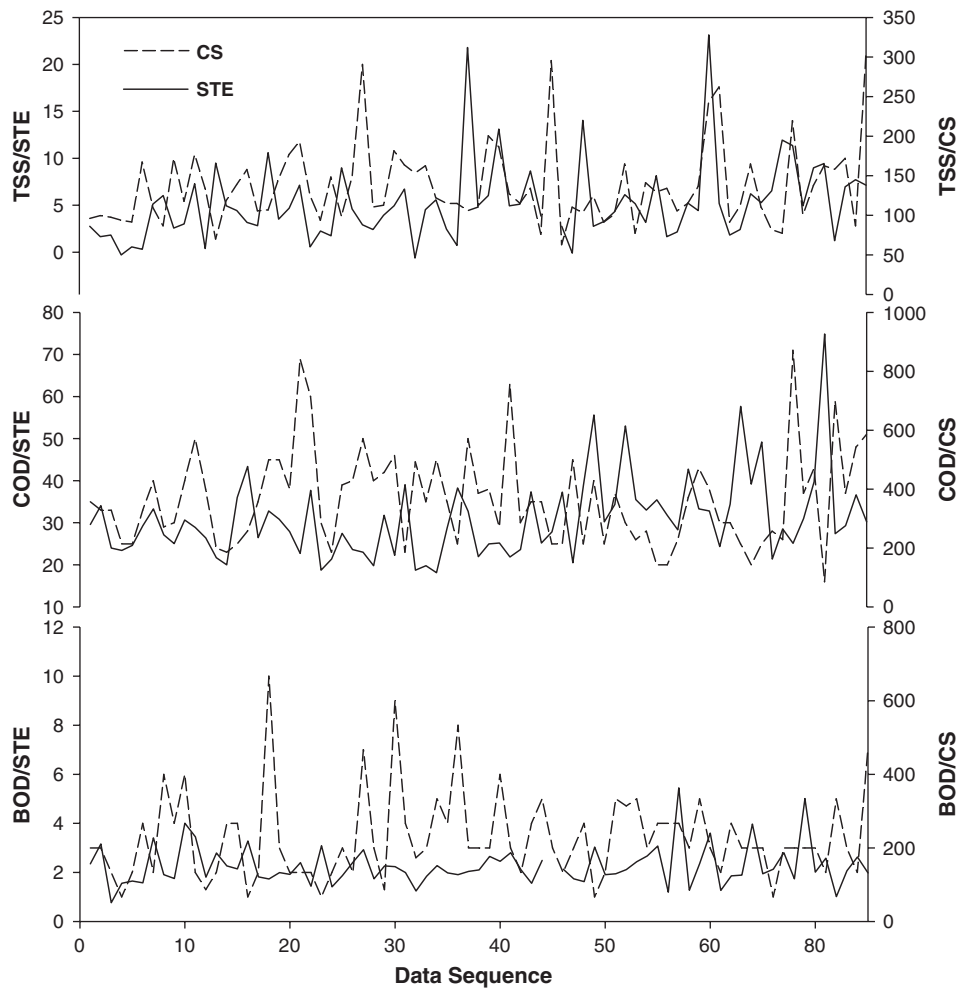


Fig. 2. Crude supply (CS) input (right scale) and secondary treatment (STE) output data (left scale) of Doha West wastewater treatment plant.

conventions BOD/CS, COD/CS, TSS/CS, BOD/STE, COD/STE, and TSS/STE hold for BOD in the crude supply, COD in the crude supply, TSS in the crude

supply, BOD in the STE stream, COD in the STE stream, and TSS in the STE stream, respectively. The BOD, COD and TSS were selected because they can be used as

measures for the effectiveness of the wastewater treatment plant.

4. Data preparation, preprocessing and statistical analysis

Data refining was performed on the raw experimental data by excluding all outliers which were unusual points. The existence of these outliers is due to many reasons such as transcription or transposition errors due to improper input of data. Other reasons include experimental errors or human errors. The data refining was accomplished by removing measurements that were not within the range of $\pm 3\sigma$, standard deviations around the group or design cell mean (Statistica Package, Version 6).

Various statistical manipulations can be performed in order to decipher trends in the data series. These are known as *smoothing techniques* and are designed to reduce or eliminate short-term volatility in the data. A smoothed series is preferred to a non-smoothed one because it can capture changes in the direction of the time series better than the unadjusted series; in addition, data smoothing eliminates the undesirable effect of possible noise in the process data. The conventional moving average technique was calculated for certain time series by consolidating the available data points into longer units of time; namely an average of several historical data points, using the following formula:

$$\bar{y}_t = \frac{y_t + y_{t-1} + \dots + y_{t-n+1}}{n}, \quad (1)$$

where y is the measured variable, t is the current time period, and n is the number of time periods in the average. In most cases, researchers use three-, four- or five-point moving averages (so that $n = 3, 4$ or 5). It should be emphasized that the larger the n , the smoother the series. In this work, the four degree moving average was used to generate smoothed data from the crude data. Thus, smoothed data series were used for predictions using the ANN model.

The last step in the data preparation procedure is the data scaling. This is a standard procedure for the neural

networks data preparation. The main objective here is to ensure that the statistical distribution of the values for each net input and output is roughly uniform. In addition, the values should be scaled to match the range of the input neurons. The data sets are usually scaled so that they always fall within a specified range or they are normalized so that they have zero mean and unity standard deviation. This is achieved by normalizing the mean and standard deviation of the data set.

The preprocessed data set was analyzed statistically by generating a *box and whiskers* (Tukey, 1977) plot for each variable. These plots summarize each variable by three components; a central line to indicate central tendency or location; a box to indicate variability around this central tendency and whiskers around the box to indicate the range of the variable. This is shown in Fig. 3 for the CS data and STE data. The plots illustrate the extent of outlier density in each variable as indicated by the points extending beyond the whiskers. In addition, it shows the range of each variable and, consequently, the efficiency of the plant treatment.

5. Neural networks modeling; background and methodology

The development of the neural networks started in the 1940s to help cognitive scientists to understand the complexity of the nervous system. They evolved steadily and were adopted in many areas of science. Basically, the ANNs are numerical structures inspired by the learning process in the human brain. They are constructed and used as alternative mathematical tools to solve a diversity of problems in the fields of system identification, forecasting, pattern recognition, classification, process control and many others (Huang and Mujumdar, 1993; Narendra and Mukhopadhyay, 1994; Rivals and Personnaz, 1996; Joaquim and Dente, 1997; Shaw et al., 1997; Baker and Richards, 2002). The interest in ANN as a mathematical modeling tool resulted in the consolidation of its theoretical background and the development of its underlying learning and optimization algorithms.

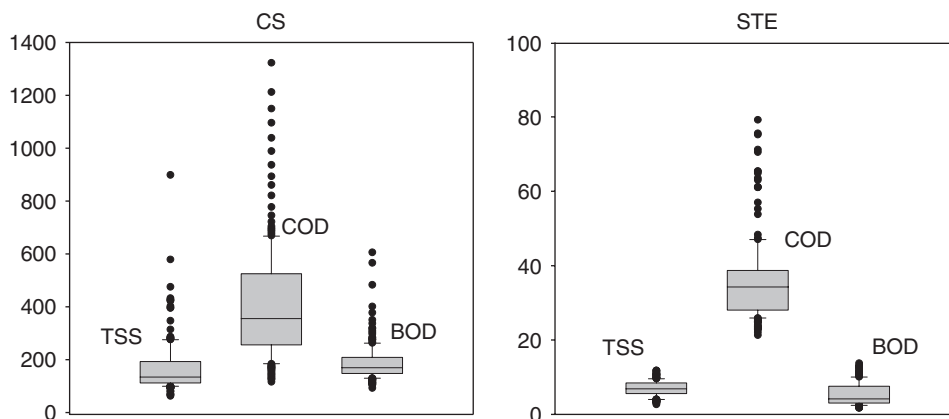


Fig. 3. Box diagrams for the plant data for the crude supply and the secondary treatment effluent streams.

One of these research areas of interest is the modeling and simulation of chemical processes. The implementation of mechanistic models that rely on fundamental material and energy balances as well as empirical correlations involves a great deal of mathematical difficulties and in many instances lacks accuracy. Neuron-based modeling can be used confidently as a substitute for such situations. This is due to the favorable features entailed in their use. Among these features are; simplicity, fault and noise tolerance, plasticity property (Shahaf and Marom, 2001) (can retain its prediction efficiency even after the removal or damage of some of its neurons), black box modeling methodology, capability to adapt to process changes.

The ANNs can be categorized in terms of topology such as single and multi-layer feedforward networks (FFNN), feedback networks (FBNN), recurrent networks (RNN), self-organized networks. In addition, they can be further categorized in terms of application, connection type and learning methods. The most commonly used type of networks in the field of modeling and prediction is the FFNN shown in Fig. 4. In this topology, the network is composed of one input layer, one output layer and a minimum of one hidden layer. The term feedforward describes the way in which the output of the FFNN is calculated from its input layer-by-layer throughout the network. In this case, the connections between network neurons do not form cycles. No matter how complex the network is, its building block is a simple structure called the neuron. It performs a weighted sum of its inputs and calculates an output using certain predefined activation functions. Activation functions for the hidden units are needed to introduce the nonlinearity into the network. The *Sigmoidal* functions, such as *logistic* and *tanh*, and the *Gaussian* function, are the most common choices for the activation functions. The neural system architecture is defined by the number of neurons and the way in which the neurons are interconnected. The network is fed with a set of input–output pairs and trained to reproduce the outputs. The training is done by adjusting the neurons weights using an optimization algorithm to minimize the quadratic error between observed data and computed outputs. A good reference on the FFNN and their applications is given by Fine (1999).

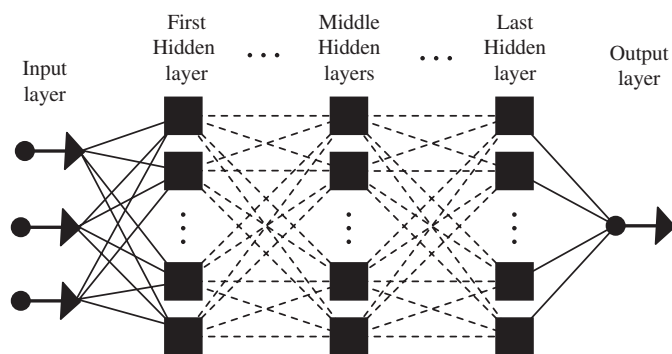


Fig. 4. Schematic of the multi-layer ANN structure.

Input-target training data are usually pretreated as explained above in order to improve the numerical condition for the optimization problem and for better behavior of the training process. Thus, the data are normally divided into three subsets; *training*, *validation* and *testing* subsets. The *training* subset data are used to accomplish the network learning and fit the network weights by minimizing an appropriate error function. *Backpropagation* is the training technique usually used for this purpose. It refers to the method for computing the gradient of the case-wise error function with respect to the weights for a feedforward network. The performance of the networks is then compared by evaluating the error function using the *validation* subset data, independently. The *testing* subset data are then used to measure the generalization of the network (i.e. how accurately the network predicts targets for inputs that are *not* in the training set) this is some times referred to as *hold-out validation*.

Improperly trained neural networks may suffer from either *underfitting* or *overfitting*. The former describes the condition when a network that is not sufficiently complex fails to fully detect the signal in a complicated data set. On the other hand, the latter condition occurs when a network that is too complex may fit the noise, in addition to the signal. This condition must be avoided because it may lead to predictions that are far beyond the range of the training data and produce wild predictions even with noise-free data. There are many reported techniques to avoid underfitting and overfitting such as model selection, jittering, early stopping, weight decay, Bayesian learning, and combining networks (Smith, 1996).

Selecting network structure is a crucial step in the overall design of NNs. The structure must be optimized to reduce computer processing, achieve good performance and avoid overfitting. The selection of the best number of hidden units depends on many factors. The size of the training set, amount of noise in the targets, complexity of the sought function to be modeled, type of activation functions used and the training algorithm all have interacting effects on the sizes of the hidden layers. There is no way to determine the best number of hidden units without training several networks and estimating the generalization error of each. If there are few hidden units, then high training error and high generalization error due to underfitting may occur. On the other hand, if many hidden units are used, low training error can be achieved at the expense of network generalization which degrades overfitting (Geman et al., 1992).

6. Results and discussion

6.1. Statistical analysis

As a preliminary multivariable statistical analysis criterion, the correlation matrix (CM) is used to explore the degree to which a linear model may describe the relationship between the variables. The CM is a table of all possible correlation coefficients between a set of variables. Each

Table 1
Correlation coefficients matrix for the plant data variables

Variable/Unit	TSS/CS	COD/CS	BOD/CS	TSS/STE	COD/STE	BOD/STE
TSS/CS	1.00	0.75	0.44	−0.02	0.40	0.13
COD/CS	0.75	1.00	0.44	−0.03	0.29	0.28
BOD/CS	0.44	0.44	1.00	0.19	0.17	0.04
TSS/STE	−0.02	−0.03	0.19	1.00	−0.08	−0.14
COD/STE	0.40	0.29	0.17	−0.08	1.00	0.07
BOD/STE	0.13	0.28	0.04	−0.14	0.07	1.00

element in this matrix is a correlation coefficient that measures the degree of linear relationship between two variables (column variable versus row variable).

The correlation coefficient (some times called the Pearson product moment correlation coefficient) is the most widely used measure of correlation or association. The correlation coefficient is described as the sum of the product of the Z -scores, Z_X and Z_Y , for the two variables divided by the number of scores

$$R = \frac{\sum Z_X Z_Y}{N}. \quad (2)$$

If the Z -scores expression is substituted into this formula, the following formula for the Pearson product moment correlation coefficient can be obtained:

$$R = \frac{\sum (X - \mu_X)(Y - \mu_Y)}{N\sigma_X\sigma_Y}, \quad (3)$$

where μ_X and μ_Y are the means of the X and Y scores, respectively; σ_X and σ_Y are the variances of the X and Y scores; and N is the number of available subjects.

The idea behind the computation of the CM is to predict one variable from the other using linear relations. It does not give any indication for any relations that divert from linearity. Nevertheless, this matrix serves as a preliminary indication of probable correlation between the data set variables.

For the data set under consideration, the CM was calculated as shown in Table 1. To measure the extent of existing correlations between the variables, the significance level of the correlation coefficients (p -level) was calculated. The p -level represents a decreasing index of the reliability of a result. The degree of confidence in the observed relation between variables decreases as the p -level becomes higher. A p -level of 0.05 is customarily treated as an acceptable probability of error (Brownlee, 1960). In Table 1, the correlations marked as bold are significant at $p < 0.05$. The calculated CM indicates some degree of linear correlations between the variables in the CS and that in the STE streams. The weakness of these correlations indicates that the use of conventional regression techniques in modeling such a complex process is irrelevant and there is a great need for using more powerful methods.

6.2. Modeling results

The previously described neural networks design procedure is applied to model the WWTP. The neural networks toolbox of the MATLAB package is utilized for this analysis. Two ANN input topologies are considered for the plant modeling. In the first approach, each of the influent variables (TSS, COD, or BOD) is used to predict each of the effluent variables. In the second approach multi-input variables are used to predict the corresponding output variables in the effluent stream.

The input–output data are grouped in two vectors (one input and one output) for the first approach and four vectors (three inputs and one output) for the second approach. The data vectors are pre-processed to fall in the range $[-1,1]$ by calculating the minimum and the maximum of each vector variable and scaling the data with respect to these limits. This is achieved by using the MATLAB function *premnmx*. This is important for improving the efficiency of network training. Each of the data set vectors was subdivided into three groups: training, validation and testing, in a ratio of 4:2:1, respectively.

Each network structure is selected after running a number of preliminary experiments to explore the training speed and response time of different structures. To keep the network structure as simple as possible, three layers are used in all single input networks (one input layer, one hidden layer and an output layer) and four layers are used for the multi-input case (two hidden layers). Results for one hidden layer in the multi-input case gave poor results. The number of neurons in the output layer is limited to the number of outputs. On the other hand, the number of neurons in the first two layers is selected after testing the performance of the networks at different combinations. It is noticed that 40 neurons is the least number of neurons, in the hidden layer, which converged to a final solution. However, for the multi-input case the two hidden layers contain 20 and 10 neurons, respectively. The prediction of TSS using the BOD as an input required 80 neurons in the hidden layer. The selected structures insure training with reasonable speed and short simulation time for a specific network performance. The constituents of the network layers, i.e. types of neurons, were taken to be *tan-sigmoidal* and *linear*. This is a common choice for function approximation neural networks (Demuth and Beale, 2000).

The optimization algorithm used for all network training runs was the *Levenberg–Marquardt*. The MATLAB routine *trainlm* with memory reduction was used for the optimization. Based on previous experience, it was found that this algorithm attains fast learning speed and high performance relative to other optimization algorithms. The details of this algorithm are reported by Hagan et al. (1996).

The performance function used for training is based on the mean square errors (MSE) between actual plant output and network predictions. Based on the selected network structure, the training process was activated to achieve a performance target of 1×10^{-3} for a maximum training epochs of 1000. The learning rate was chosen to be 0.01. The value of this parameter was obtained after performing several trial and error runs. It was found that this value

insures stable fast learning. In this work, the time required for the networks training was in the range 2–5 min for the case of single input networks, whereas it required 10–20 min for the three multi-input networks. This is due to the structure complexity of the multi-input networks. The early network training termination technique was used to assure network generalization and to prevent over- or under-fitting. The ANN predictions for the Doha West WWTP are shown in Figs. 5–7.

The NN prediction results from the STE outputs using only a single input set of data for the crude supply are shown in Fig. 5. It is noticed that the use of COD as an input in the CS supply resulted in more adequate predictions of the BOD, COD and TSS in the STE stream than that when using TSS as an input in the CS stream.

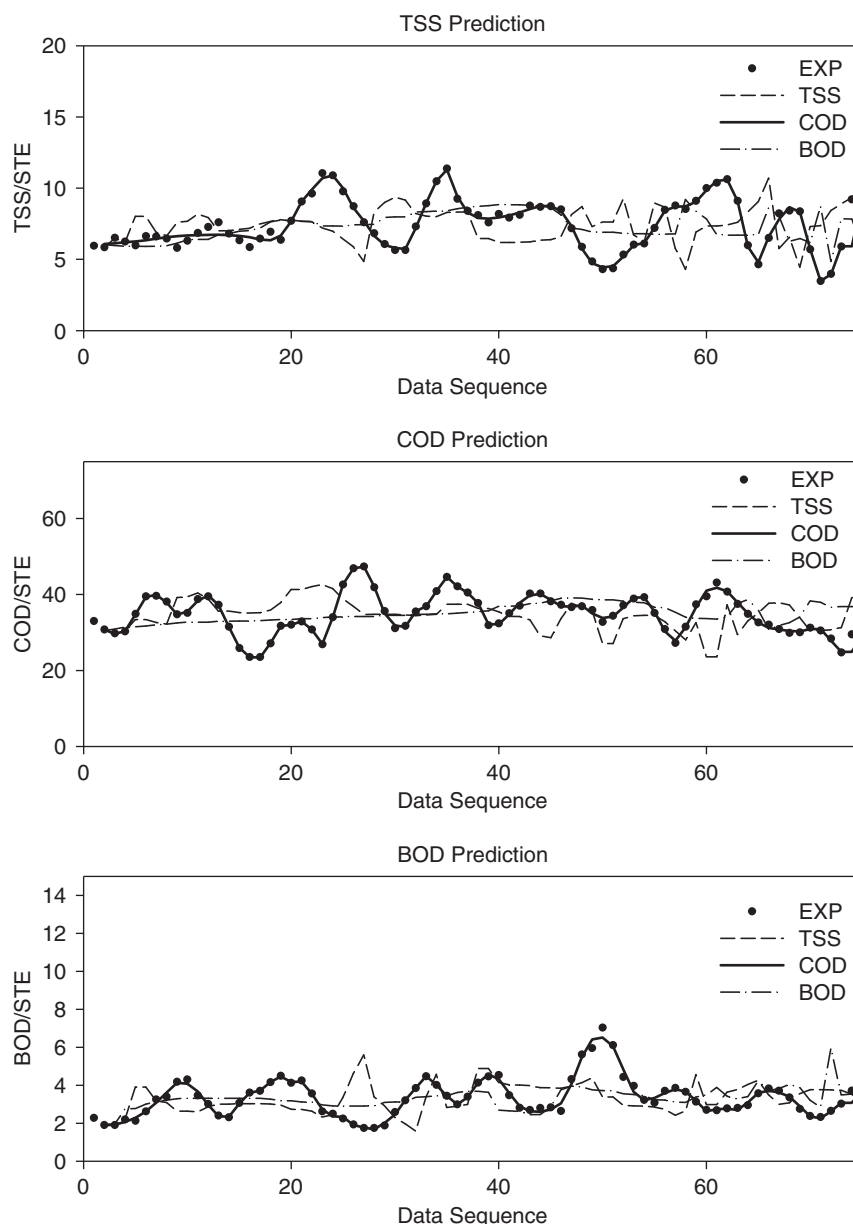


Fig. 5. ANN predictions for the TSS, COD, and BOD variables from the STE stream as a function of variables in the CS stream.

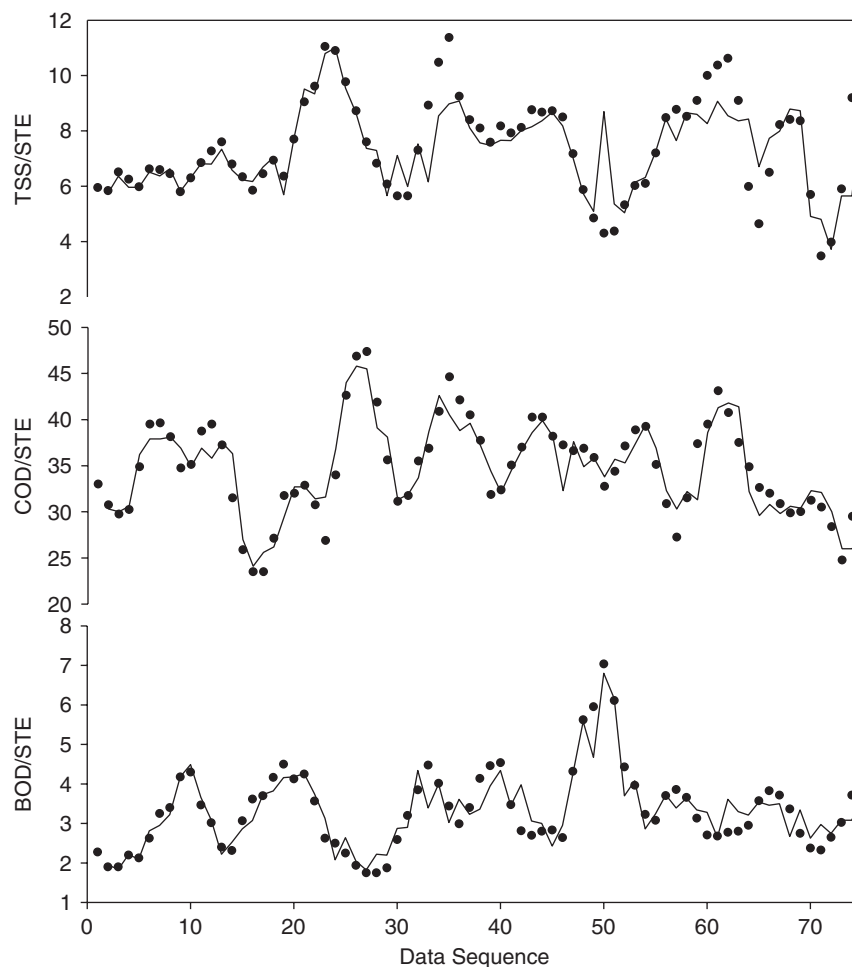
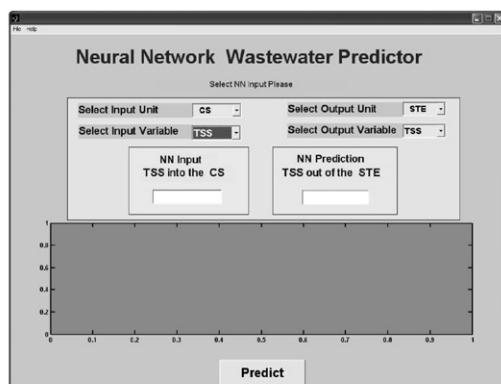


Fig. 6. TSS, COD and BOD ANN predictions for the case of three CS stream inputs and one STE stream output variable.

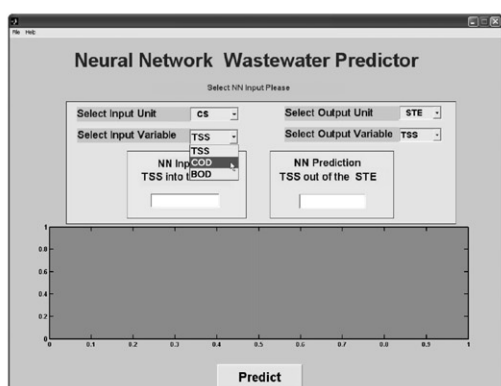
Poor predictions of the outputs were obtained when using BOD as an input in the CS stream. These conclusions are supported by the values of MSE and R (an indication of goodness-of-fit) which are displayed in Table 2. It is seen that MSE values are the least and R values are close to 1.0, when COD/CS is used as a feed input for the ANN predictions of the outputs.

Fig. 6 shows results when all of the input variables, BOD, COD and TSS, in the CS stream are used at once as feed to the NN to predict BOD, COD and TSS in the STE stream. The predictions are highly satisfactory; however, they are still slightly incomparable to single input predictions using COD/CS, but much better than the single input predictions using TSS/CD or BOD/CS as feed inputs to the neural network. This is due to the influence, or interactions, of the feed input variables on each other. In other words, when BOD/CS alone was used for output predictions it resulted in inadequate ANN predictions, but when COD/CS alone was used for output predictions it resulted in adequate predictions; thus, when both BOD/CS and COD/CS are used a decline in the output predictions would be expected.

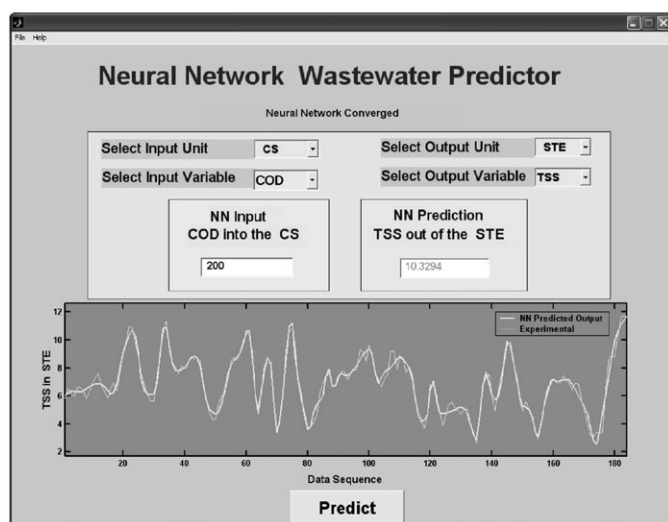
In order to facilitate the use of the trained ANNs and to eliminate the chance of improper use of the network models, a graphical user interface (GUI) was written using the MATLAB mathematical scripting language. Fig. 7 shows three screen captures of the program execution. The GUI main window provides the user with four pull down menus; two for the input unit and variable selection, with the other two for the output unit and variable selection. The available units are CS, primary treatment effluent (PTE), and STE and the input–output variables are the TSS, COD and BOD. The graphical representation of the corresponding ANN output prediction as well as the plant experimental data are displayed and updated automatically upon the selection of any input or output units and the desired variables. To calculate the ANN prediction for a particular plant input, the user should enter the value in the ANN input field and press a designated prediction button. The input value is processed internally by scaling, network simulation and de-scaling. A proper warning message is displayed to inform the user and give the acceptable prediction range in case an input value lies outside the ANN prediction range. The user has the choice of



Initial screen: Program startup window



Entry of Input: I/O units and variables as well as NN input value



Final Screen: Prediction results and NN selected profile

Fig. 7. Graphical user interface representation of the ANN predictions for the Doha West WWTP.

predicting any plant output using different plant input variables. Fig. 7 shows the main steps of running the GUI program for a typical set of input–output data.

7. Conclusions

Modeling a WWTP is difficult to accomplish due to the high nonlinearity of the plant and the non-uniformity and variability of the crude supply as well as the nature of the

Table 2

Summary of trained ANN results for different input–output variables combinations

Input variable in CS	Output variable in STE	NN structure	MSE	R
TSS	TSS	1-40-1	0.146	0.854
COD	TSS	1-40-1	0.021	0.987
BOD	TSS	1-80-1	0.224	0.748
TSS	COD	1-40-1	0.047	0.735
COD	COD	1-40-1	0.014	0.923
BOD	COD	1-40-1	0.304	0.634
TSS	BOD	1-40-1	0.255	0.785
COD	BOD	1-40-1	0.061	0.951
BOD	BOD	1-40-1	0.366	0.568
TSS, COD, BOD	TSS	1-20-10-1	0.053	0.839
TSS, COD, BOD	COD	1-20-10-1	0.030	0.924
TSS, COD, BOD	BOD	1-20-10-1	0.023	0.924

biological treatment. An ANN modeling approach was implemented to solve this problem and to discover the interdependency of input–output variables. The plant input–output data were used to predict the plant behavior without using mechanistic bio-modeling which involves a great degree of complexity and uncertainty.

The modeling approaches used in this study, namely single input and multi-input networks topologies, gave comparable predictions of the plant performance criteria. The first approach gave better predictions when the COD was used as a network input. On the other hand, the second approach gives reasonable results for all predicted variables. The ANN modeling technique has many favorable features such as efficiency, generalization and simplicity, which makes it an attractive choice for modeling complex systems, such as wastewater treatment processes.

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